Astronomy A1: The Night Sky and Celestial Motion

MAIS Physics & Astronomy Club (PAC)

Before pointing a telescope upward, every astronomer learns to read the celestial sphere with the naked eye: measuring angles on the sky, predicting which stars rise or stay visible all night from our latitude in Ulaanbaatar, and understanding how Earth's spin and orbit shape days, months, and seasons.

1. Angles on the Sky and Parallax

Looking at the sky alone tells us directions, not distances. One degree (1°) is subdivided into 60 arcminutes (60′), and each arcminute into 60 arcseconds (60″), so 1 rad = 180°/π ≈ 206,265″. Whenever an object of linear diameter **d** sits at a much larger distance **D** ≫ **d**, its angular diameter is

$$\theta \text{ (in radians)} \approx (\mathbf{d})/(\mathbf{D}),\quad \theta \text{ (in arcseconds)} \approx 206{,}265\ (\mathbf{d})/(\mathbf{D}). \tag{1}$$

Annual parallax uses Earth's orbital radius, 1 AU, as the baseline: a nearby star's parallax angle is **p** = (1 AU)/**D**, half the peak-to-peak shift against distant background stars over six months. By definition, a star at distance **D** = 1 parsec (pc) has parallax **p** = 1″, so **1 pc = 206,265 AU ≈ 3.26 light-years**.

[1] Problem 1 (A cosmic coincidence). The Sun has radius $\mathbf{R}_{\odot} \approx 6.96 \times 10^5$ km at mean distance $\mathbf{D}_{\odot} \approx 1.50 \times 10^8$ km, while the Moon has radius $\mathbf{R}_{\mathbf{M}} \approx 1{,}737$ km at mean distance $\mathbf{D}_{\mathbf{M}} \approx 3.84 \times 10^5$ km.

(a) Calculate the mean angular diameters $\theta_{\odot}$ and $\theta_{\mathbf{M}}$ in arcminutes.

(b) Because the Moon's orbit is slightly elliptical (**e** ≈ 0.055), some central solar eclipses are **total** while others are **annular** ("ring of fire"). Explain why, and explain why solar eclipses do not occur at every new moon.

2. The Celestial Sphere from Ulaanbaatar

Projecting Earth's equator and poles onto the sky defines the **celestial equator** (declination δ = 0°) and the north/south celestial poles (δ = ±90°). For an observer at northern geographic latitude φ (for Ulaanbaatar, **φ ≈ 47.9° N**):

- The north celestial pole (marked closely by Polaris) sits due north at altitude $\mathbf{h}_{\mathrm{NCP}} = \varphi$ above the horizon.
- A star of declination δ reaches its highest altitude (upper culmination) when crossing the meridian:

$$\mathbf{h}_{\max} = 90° - |\varphi - \delta|. \tag{2}$$

Stars with $\delta > 90° - \varphi$ never dip below the northern horizon (**circumpolar**), whereas stars with $\delta < -(90° - \varphi)$ never rise above the southern horizon.

Remark. Check two extreme latitudes in equation (2): at the North Pole ($\varphi = 90°$), Polaris is straight overhead ($\mathbf{h} = 90°$) and the northern sky never sets; at the equator ($\varphi = 0°$), every star on the celestial sphere spends 12 hours above the horizon and 12 hours below.

[2] Problem 2 (Stargazing over Mongolia). Consider observing from Ulaanbaatar at latitude $\varphi = 47.9°$ N (neglecting atmospheric refraction and horizon hills).

(a) Find the declination range of stars that are circumpolar from Ulaanbaatar, and the declination range of stars that can never be seen.

(b) Classify each of the following bright stars—Dubhe in Ursa Major ($\delta = +61.8°$), Vega ($\delta = +38.8°$), Sirius ($\delta = -16.7°$), and Alpha Centauri ($\delta = -60.8°$)—as circumpolar, seasonal (rises and sets), or never visible, and find both the maximum and minimum altitudes of Dubhe and the maximum altitude of Sirius.

3. Solar Time, Sidereal Time, and Synodic Periods

Because Earth orbits the Sun in the same sense that it spins on its axis, after one 360° rotation relative to the distant stars (one **sidereal day**) the Sun has shifted eastward along the ecliptic by nearly 1°, so Earth must turn an extra ~4 minutes to bring the Sun back to the meridian (one 24-hour **solar day**).

[2] Problem 3. Earth completes one orbit around the Sun in $\mathbf{N} = 365.2422$ mean solar days.

(a) Explain geometrically why Earth completes $\mathbf{N} + 1$ (not $\mathbf{N}$) rotations relative to the distant stars in one year, and calculate the length of a sidereal day in hours, minutes, and seconds.

(b) If Orion crosses the southern meridian at 23:00 local time tonight in Ulaanbaatar, at what clock time will Orion cross the meridian 30 days later?

[3] Problem 4 (Synodic periods of the Moon and planets). Two bodies orbit the same central mass in circular coplanar orbits in the same direction with sidereal periods $\mathbf{P}_1 < \mathbf{P}_2$. Their **synodic period S** is the time interval between successive alignments on the same side of the center.

(a) By comparing their angular speeds $\omega_{\mathbf{i}} = 2\pi/\mathbf{P}_{\mathbf{i}}$, prove that $(1)/(\mathbf{S}) = (1)/(\mathbf{P}_1) - (1)/(\mathbf{P}_2)$.

(b) The Moon orbits Earth relative to the stars in $\mathbf{P}_{\mathbf{M}} = 27.32$ days while Earth orbits the Sun in $\mathbf{P}_{\oplus} = 365.25$ days. Compute the cycle of lunar phases (new moon to new moon).

(c) Explain why distant Neptune ($\mathbf{P_N} \approx 165$ yr) reaches opposition from Earth almost every year ($\mathbf{S} \approx 367.5$ d), whereas nearby Mars ($\mathbf{P}_{\text{Mars}} \approx 1.88$ yr) reaches opposition only once every ~780 days!